

14.	C	K.E. of a molecule = $\frac{1}{2} m v_{\text{rms}}^2$ $KE = \frac{3}{2} kT$ T (in K) and KE are proportional to v_{rms}^2 When v increases to 2v New KE = $\frac{1}{2} m (2v)^2 = \frac{1}{2} m v^2 \times 4 = 1.6 \times 10^{-23} \times 4 = 6.4 \times 10^{-23} \text{ J}$ New T = $573 \times 4 = 2292 \text{ K}$
15	D	Bob has maximum PE at $t = 0$ means that object starts at either extreme